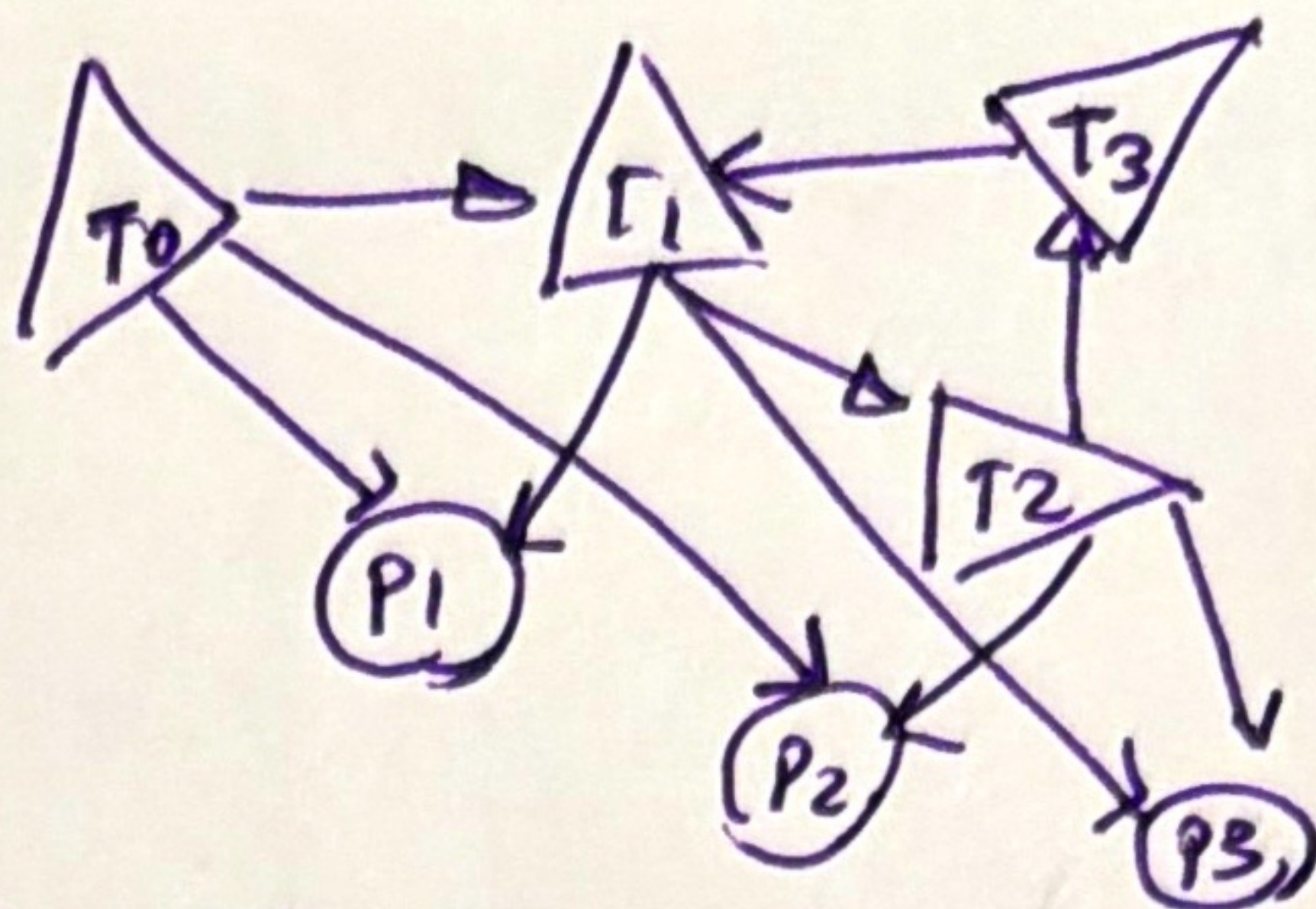


Quiz - Solution
CS 4733 15 733
Fall 2025

a)

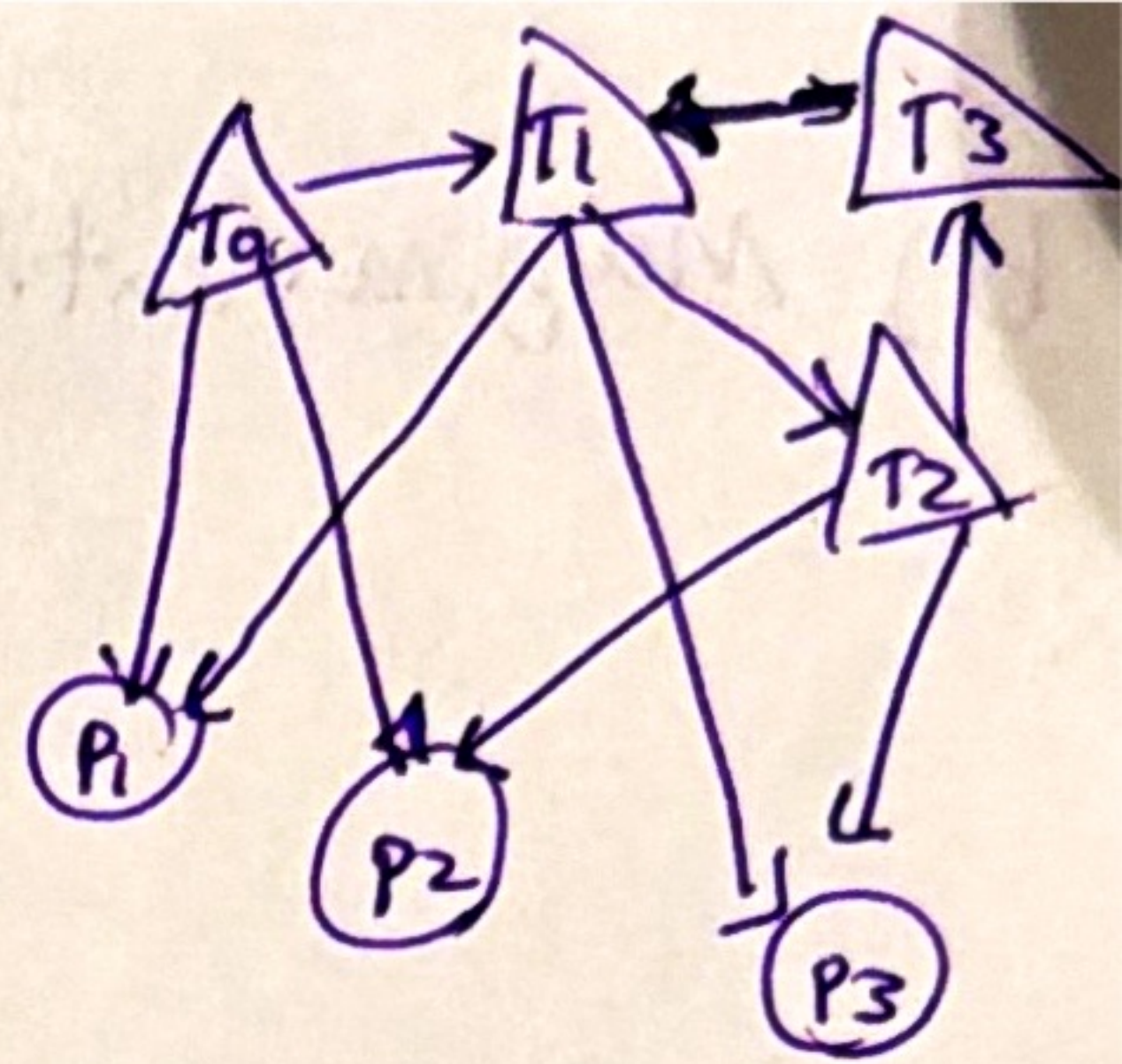


Jacobian (J)

	T_0	T_1	T_2	T_3	P_1	P_2	P_3
T_0	1	1					
T_1		1	1				
T_2			1	1			
T_3				1			
$T_0 P_1$	1				1		
$T_0 P_2$	1					1	
$T_1 P_1$		1			1		
$T_1 P_3$		1					1
$T_2 P_2$			1			1	
$T_2 P_3$			1				1

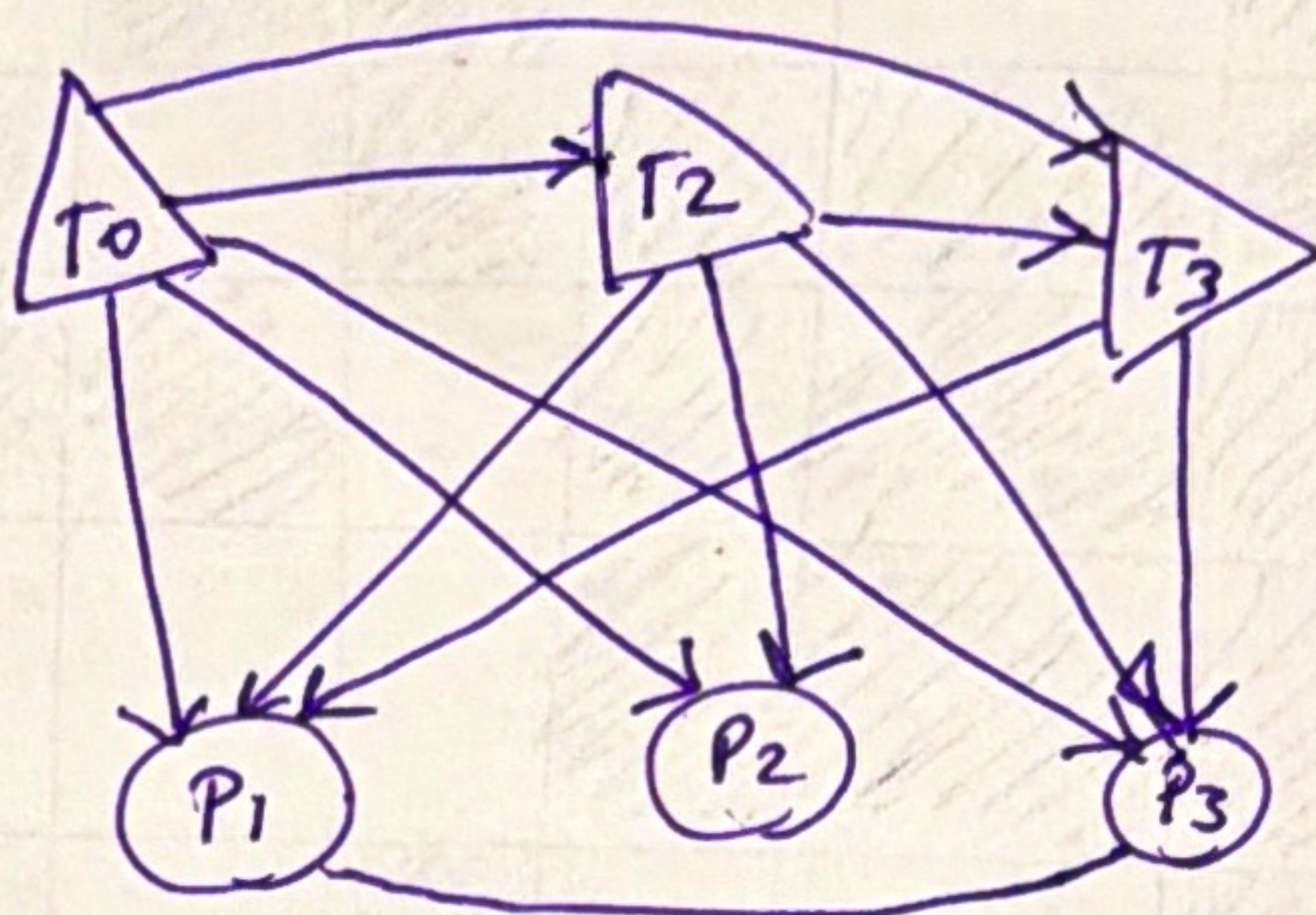
Hessian $(\gamma^T \gamma)$

	T_0	T_1	T_2	T_3	P_1	P_2	P_3
T_0							
T_1							
T_2							
T_3							
P_1							
P_2							
P_3							



b) Marginalization around T_1

New pose graph:



Jacobian for marginalized graph (J)

	T_0	T_2	T_3	P_1	P_2	P_3
T_{02}						
T_{23}						
T_{03}						
$T_0 P_1$						
$T_0 P_2$						
$T_0 P_3$						
$T_2 P_1$						
$T_2 P_2$						
$T_2 P_3$						
$T_3 P_1$						
$T_3 P_2$						
$T_3 P_3$						
$P_1 P_3$						

Hessian ($J^T J$) for marginalized graph

	T_0	T_2	t_3	P_1	P_2	P_3
T_0	///	///	///	///	///	///
T_2	///	///	///	///	///	///
T_3	///	///	///	///		///
P_1	///	///	///	///		///
P_2	///	///			///	
P_3	///	///	///	///		///

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